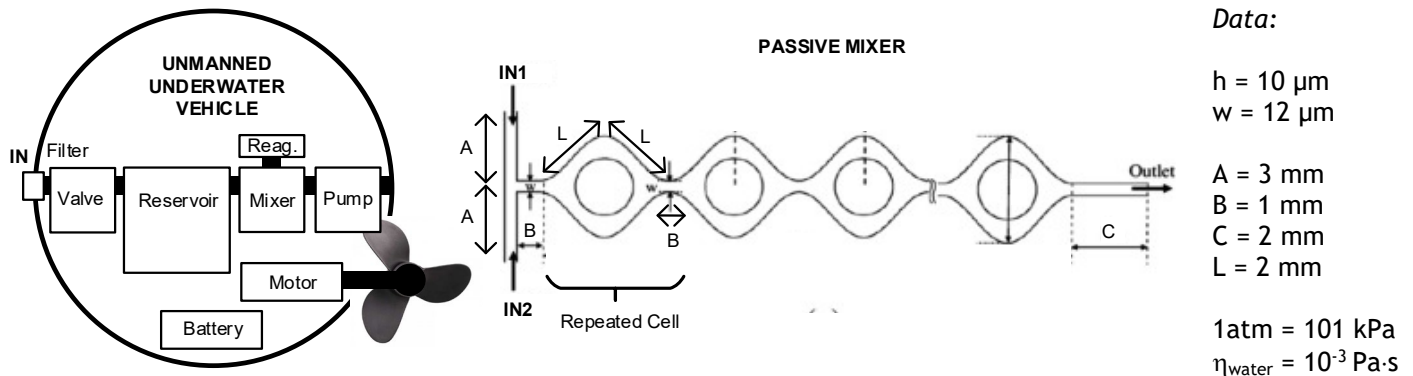


Politecnico di Milano
Biochip A.Y. 2020/2021 - M. Carminati
 Exam of 07.09.2021

*Test duration: 2 hours - No support material (textbooks, laptops, smartphones, notes) admitted.
 All the answers must be motivated. Beyond the final numerical results, all the expressions must be provided.*

Exercise 1

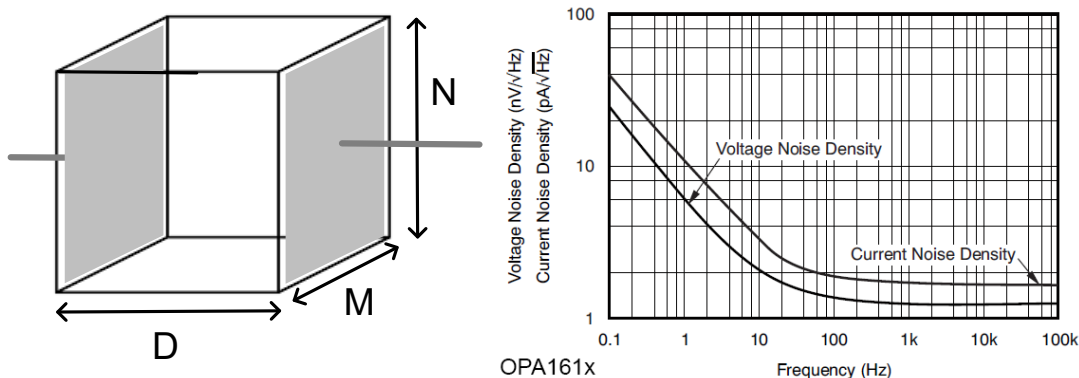
Consider an underwater drone (a sphere of 20 cm diameter) for wireless monitoring water quality in lakes and rivers. It contains a microfluidic system to analyze the water while navigating, driven by a pump working in aspiration. The filling level of the water reservoir is also used to control vertical buoyancy. The passive mixer has a periodic structure of a repeated cell unit where channel height is h and assume constant width w in all segments.



- Draw the electrical fluidic equivalent the mixer.
- Assuming 4 repeated “diamond” cells and $P_{\text{IN1}} = 2 \text{ atm}$, $P_{\text{IN2}} = 1.5 \text{ atm}$ and 1.2 atm at the mixer outlet, find the water speed in all sections of the mixer.
- If the drone is travelling in the water at a horizontal speed of 1 cm/s , what is the amount of the drag force?
- Assuming that the average density of the drone is 3 times that of water, what fraction of the total sphere volume should be the reservoir volume in order to allow buoyancy (when fully empty)?
- What should be the cutoff size (max. size allowed in filter pores) of the input filter?

Exercise 2

Now focus on the reservoir. A couple of flat parallel gold electrodes on the sides of the reservoir in contact with the liquid (see figure below) allow to measure both the level (capacitive) as well as the conductivity of the water.



- Find the equivalent impedance between the electrodes for both empty and full reservoir assuming a water conductivity equal $1/100$ of that of PBS and a stray capacitance between electrodes of 1 pF due to connections.
- Design two separate circuits for the measurement of level (from empty to full) and of water conductivity (consider full reservoir) and chose the values of their parameters.
- Considering that the attitude control systems requires a max. time constant of 1 ms , choose the bandwidth of the block measuring the reservoir level.
- Assuming the empty reservoir connected to a transimpedance amplifier with $1 \text{ k}\Omega$ feedback resistor and opamp OPA161 (its input-referred noise equivalent generators whose spectra are reported in the figure) find the total input noise spectrum.
- What would change if the metal of the electrodes were changed to Platinum or Copper?